

IUFD(INTRA UTERINE FETAL DEATH)



- **Definition :**

- Intrauterine fetal death (IUD) embraces all fetal death weighing 500 gm or more occurring both during pregnancy (antepartum death) or during labour (intrapartum).

- Thus, ***Antepartum death occurring beyond the period of viability is termed as intrauterine death.***

- **WHO definition:**

- Fetal death means death prior to complete expulsion or extraction of mother of a fetus irrespective of duration of pregnancy and which is not an induced termination pregnancy.



- **ETIOLOGY:**

- *Causes*

- Maternal: 5-10%
- Placental : 20-35%
- Fetal : 25-40%
- Unknown: 25-35%



1. Maternal 5-10%

- Hypertensive disorder in pregnancy.
- Diabetes in pregnancy.
- Infections(malaria,hepatitis, influenza, toxoplasma&syphili
- Hyperpyrexia
- Antiphospholipid syndrome
- Thrombophilis
- Abnormal labour
- Post term pregnancy
- Systemic lupus erythematosus.

2. Fetal 25-40%

- Chromosomal abnormalities
- Major structural anomalies
- Infections
- Rh- incompatibility
- Non immune hydrops
- Growth restriction



3. **Placental** 20-35%

- Antepartum haemorrhage.
- Cord accident
- Placental insufficiency
- Twin transfusion syndrome (TTTS)

4. **Iatrogenic**

- Beyond dose.
- External cephalic version
- Drugs.

5. **Idiopathic** 25-35%

- Causes remains unknown even with thorough clinical examination and investigations.



DIGNOSIS:

- Repeated examination are often required to confirm the diagnosis.
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SYMPTOMS:

- Absence of fetal movements which were previously noted by the patient.



SIGNS: retrogression of the positive breast changes that occur during pregnancy is evident after variable period following death of the fetus.

- **Per abdomen:**
 - ***Gradual retrogression*** of the fundal height and it becomes smaller than the period of amenorrhoea.
 - ***Uterine tone*** is diminished and the uterus feels flaccid. Braxton-Hicks contraction is not easily felt.
 - Fetal movements are not felt during palpation
 - ***.Fetal heart sound*** is absent. Doppler ultrasound is better than the stethoscope.
 - ***Egg-shell crackling*** feel of the fetal head is late feature.
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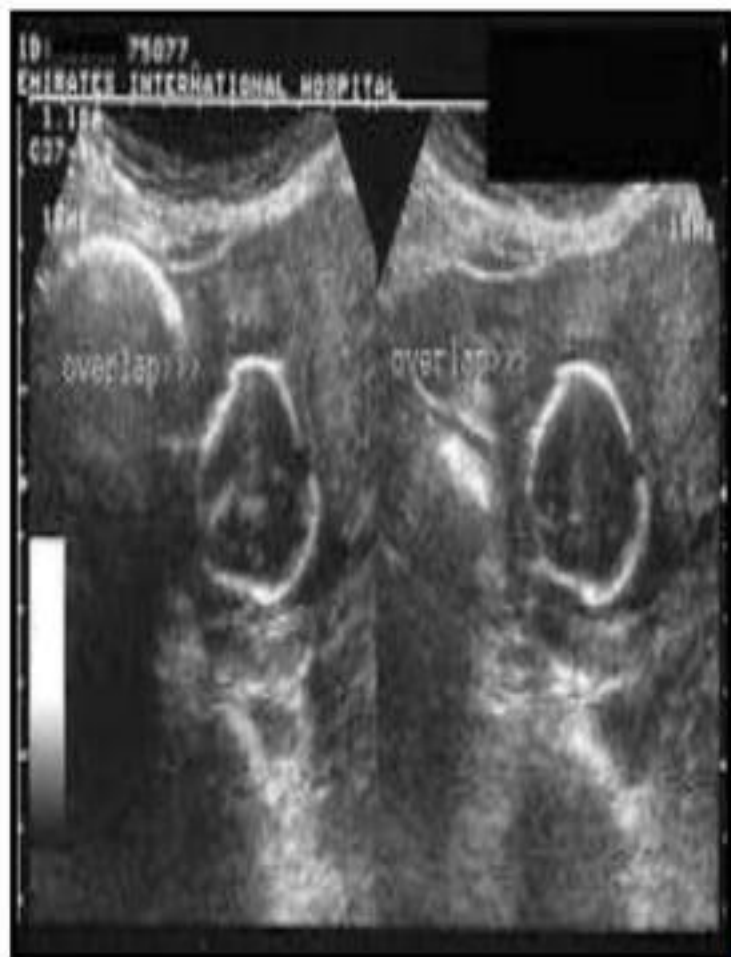
○ INVESTIGATION:

Sonography:

- Earliest diagnosis is possible with Sonography.
 - The evidences are:
 - Lack of all fetal motions during a 10minute period of careful observation with a real-time sonar is a strong presumptive evidence of fetal death.
 - Gradually, oligohydramnios and collapsed cranial bones are evident.
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- **Straight X-ray abdomen**: Rarely done at present .
The following features may be found in varying degree either singly or in combination.
- ***Spalding sign*** –
- The irregular overlapping of the cranial bones on one another is due to liquefaction of the brain matter & softening of the ligamentous structures supporting the vault. ***It usually appears 7 days after death.***
- Similar features may be found in extra-uterine pregnancy with the fetus alive





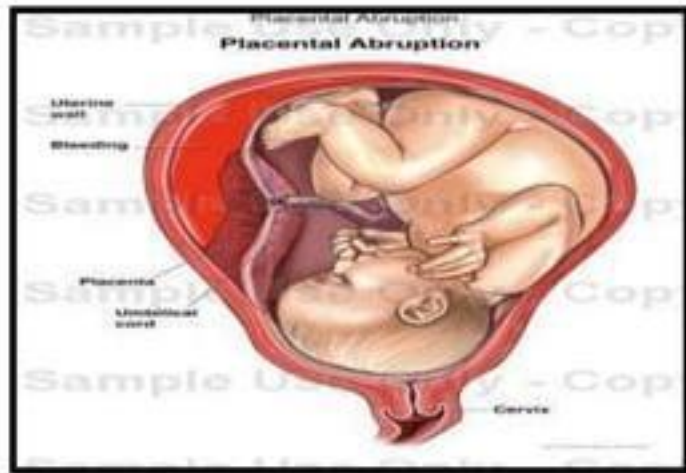
- **Hyperflexion** of the spine is more common. In some cases hyperextension of the neck is seen.
- **Crowding of the ribs shadow** with loss of normal parallelism.
- **Roberts sign : Appearance of gas shadow** in the chambers of the heart and great vessels may appear as 12 hours but difficult to interpret. When detected provides conclusive evidence.
- **Blood:** To estimate the blood **fibrinogen** level and partial **thromboplastin** time periodically, when the fetus is retained for more than 2 weeks



- **COMPLICATION:**
- **Psychological Upset** often becomes a problem.
- **Infection:** so long as the membranes rupture infection especially by gas forming organisms like *Cl.welchii* may occur. The dead tissue favours their growth with disastrous consequences.
- **Blood coagulation disorders are rare.** If the fetus is retained for more than 4 weeks(10- 20%),there is a possibility of defibrination from silent disseminated intravascular coagulopathy (DIC).It is due to gradual absorption of thromboplastin, liberated from the dead placenta and decidua, into the maternal circulation.



- **During labor:** Uterine inertia, retained placenta and postpartum haemorrhage.
- **PPH** (postpartum hemorrhage)
- **Placental abruption**



- **Shock, renal failure**
- **Sepsis**
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- **Maternal death**
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○ MANAGEMENT

Prevention : the overall risk of recurrence of still birth varies between 0-8%. The conditions that run the risks of recurrence are:

- hereditary disorders
- diabetes
- hypertension
- thrombophilias
- placental abruption
- fetal congenital malformation



- **pre-conceptional counselling and care.**
- Pre-natal diagnosis
- To screen the ***“at-risk mothers”*** during antenatal care. Carefull assessment of fetal well being and to terminate pregnancy with the earliest evidence of fetal compromise.



- **EXPECTANT ATTITUDE NON INTERFERENCE:**

- In about 80% of cases, spontaneous expulsion occurs within 2 weeks of death.
- *Fibrinogen estimation should be done weekly.*

- **REASONS FOR EARLY DELIVERY:**

- Reliable diagnosis could be made with real time ultrasonography quickly.
- Prostaglandins are available for effective induction.
- Complication should be avoided.



- **INDICATIONS OF EARLY INTERFERE:**

- Psychological upset of the patient.
- Manifestation of uterine infection
- Tendency of prolongation of pregnancy beyond 2 weeks.
- Falling fibrinogen level.

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- **METHODS OF DELIVERY:** The delivery should always be done by medical induction.

- **OXYTOCIN INFUSION** :Very effective in cases where the **cervix is favourable**

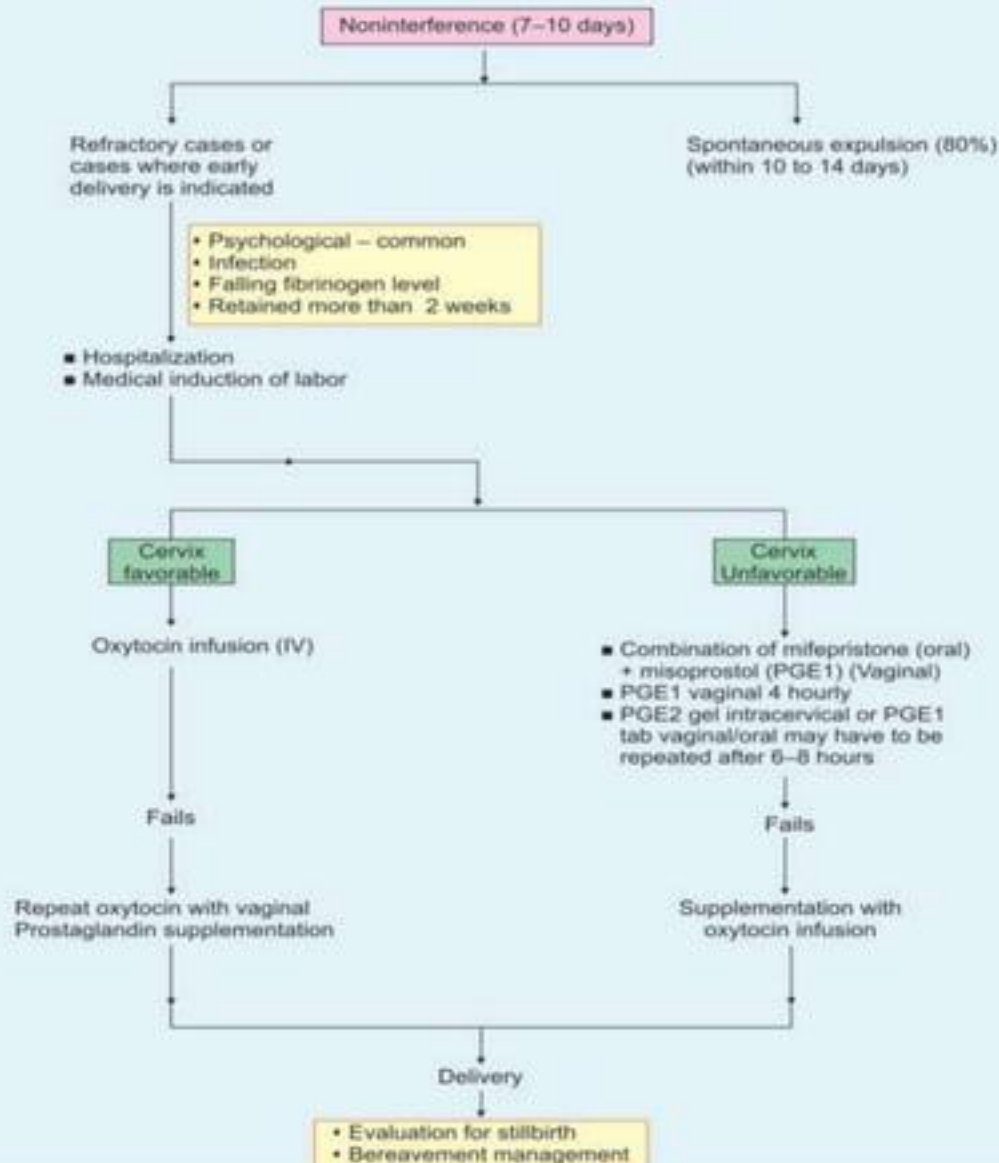
- *5-10units with 500ml of Ringer's solution is administered through intravenous infusion drip.
- * 20units with 500ml of Ringer's solution and run with 30drops in case of failure
- *if the uterus still remains refractory, the same procedure is repeated after vaginal administration of prostaglandin gel.



- **Prostaglandins:** Vaginal administration of prostaglandin (PGE₂) gel or lipid pessary high in the posterior fornix is very effective for induction where the cervix is unfavourable .
- It is repeated after 6-8 hours and may be supplemented with oxytocin infusion.
- **Misoprostol (PGE₁):**
- 25-50μ either vaginally or orally is also found effective.
- Vaginal route use is more effective compared to oral route. May be repeated for every 4 hours.



SCHEME OF MANAGEMENT OF IUFD



o THANK YOU

